

TOM OBANDE ONYANGO

Software Developer | Full-Stack and Applied AI

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PROFESSIONAL SUMMARY

Bachelor of Information Technology candidate at Kenyatta University, awaiting graduation, with 3+ years of project experience building full-stack web applications and applied-AI systems - retrieval-augmented search with source citations, deterministic financial checks and computer-vision pipelines - shipped with automated tests and continuous integration. Practical exposure to digital customer journeys through a web developer bootcamp engagement with mTek Insurance Broker. Interested in software engineering, financial technology, digital banking and data-driven service improvement.

CORE SKILLS

Languages: Python, JavaScript, TypeScript, Java, SQL

Frameworks: FastAPI, Next.js, React, Node.js, Express, SQLAlchemy, Tailwind CSS

Data and AI: PostgreSQL, Supabase, Firestore, Pinecone, RAG pipelines, embeddings, vector search, pandas, scikit-learn, MediaPipe, OpenCV

Tools: Git, GitHub, GitHub Actions, Docker, REST APIs, JWT authentication, Vercel, Netlify

Business and professional: Customer onboarding, requirements gathering, documentation, digital service delivery, risk-aware data handling, communication, teamwork, analytical thinking, problem solving

Spoken languages: English, Kiswahili, Basic Japanese

PROFESSIONAL EXPERIENCE

Web Developer Bootcamp Participant / Trainee Developer

mTek Insurance Broker | Nairobi, Kenya | One-week engagement

- Supported development of website user onboarding, contributing to a smoother sign-up and customer information capture experience.
- Translated business requirements into user-facing web features in a fast-paced developer selection bootcamp, gaining exposure to digital insurance processes and secure data capture.

IT Attache

Jaramogi Oginga Odinga University of Science and Technology | May 2024 - August 2024

- Configured and supported local area networks, improving office connectivity and day-to-day system access.
- Performed system updates, antivirus checks and software maintenance to support reliable computer operations.
- Provided end-user technical support by troubleshooting hardware, software and connectivity issues, and supported ICT team tasks requiring documentation and timely resolution of service requests.

PROJECTS

ShambaLens AI github.com/tbrowns/shambalens-ai

Evidence-first crop triage: ranks up to three competing causes from a photo rather than forcing one label, then verifies independently before presenting a plan. FastAPI, Next.js and PostgreSQL; 19 automated tests, CI against live PostgreSQL.

Mindbase github.com/tbrowns/mind-base

Internal knowledge assistant answering with source citations. Masks personal data, indexes to Pinecone, filters retrieved chunks for relevance before generation. Next.js, Firestore, Groq.

DRIP Orchestrator github.com/tbrowns/drip_orch_platform

Dividend reinvestment simulator for the Nairobi Securities Exchange: scrapes live NSE quotes, tracks dividend history, models reinvestment over time. FastAPI, SQLAlchemy, JWT auth.

Finly github.com/tbrowns/finly_platform

AI financial controller. Normalises monthly Zoho books into PostgreSQL, then runs deterministic checks for cash-flow risk, fraud indicators and reconciliation; the model explains but never determines. FastAPI, Neon PostgreSQL, Groq.

Gesture World github.com/tbrowns/gesture-world

Sign language recognition with speech synthesis: MediaPipe hand-landmark detection, scikit-learn classification, Azure Cognitive Services output.

EDUCATION

Bachelor of Information Technology

Kenyatta University | 2021 - 2026 | Awaiting graduation

Relevant areas: software development, databases, computer networks, information systems, data analysis and web technologies. Additional study: Japanese.

Kenya Certificate of Secondary Education

St. Mary's School, Yala | 2017 - 2021 | Grade B+

CERTIFICATIONS

IBM Data Science Practitioner | IBM Enterprise Design Thinking Practitioner